

Emily Tan

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Education

University of California, Riverside

Riverside, California

Bachelor of Science in Computer Science (University Honors Program)

Expected 2029

Technical Skills

Languages: C++, Python, TypeScript, JavaScript

Web Development: React, Next.js, Node.js, HTML/CSS, Tailwind CSS

Embedded: STM32, CAN Bus, ADC, DMA, IoT, Arduino, Raspberry Pi

Data Science: pandas, NumPy, Matplotlib, scikit-learn

Developer Tools: Git, GitHub, VS Code, Continuous Integration, Agile

Activities

Cutie Hack Lead Software Engineer | Cutie Hack GitHub

Apr. 2026 – Present

Association for Computing Machinery @UCR

Riverside, California

- * Building software for Cutie Hack, an ACM hackathon serving 300+ participants
- * Coordinating with software and UI/UX leads to ensure development is timely and high-quality

ACM Spark Web Developer | Rides | SHPE | AEP

Jan. 2026 – Present

Association for Computing Machinery @UCR

Riverside, California

- * Collaborated in an Agile, GitHub issue-based workflow to deliver weekly frontend features from Figma designs
- * Developed UI components using Next.js, TypeScript, and Tailwind CSS for production deployment
- * Implemented responsive design patterns to ensure consistent performance across devices and screen sizes

Computer Science Associate

Oct. 2025 – Present

Formula SAE Highlander Racing @UCR

Riverside, California

- * Redesigned control logic using state machines and DMA to improve system speed and reduce CPU load
- * Programmed STM32 microcontroller to transmit and process CAN bus messages for real-time vehicle systems

Volunteer

Oct. 2021 – Jun. 2025

High School Youth Action Team

- * Collaborated with a team of 20+ volunteers to organize and run Relay For Life fundraising events, raising over \$10,000 to support cancer patients
- * Guided 30+ elementary school students one-on-one in using technology, assisting with device navigation, Wi-Fi setup, and introductory Scratch activities
- * Provided directions, answered questions, and resolved visitor concerns for guests at the Great Park Visitor Center

Projects

Mount Cryptid | C++, UML, GoogleTest, CI | GitHub

Aug. 2026

- * Designed UML diagrams to establish game architecture and class relationships
- * Collaborated in a 4-person team using an Agile Git workflow with feature branches and pull requests
- * Developed and tested game features with GoogleTest and CI to maintain code quality and catch regressions

UCR Registration Extension | Chrome Extension, JavaScript, APIs | GitHub

Webstore

- * Built a Chrome extension that aggregates course data from multiple sources into a unified registration interface
- * Integrated MyClassGrades, UCR course difficulty data, and RateMyProfessors into a structured data pipeline
- * Implemented SPA-aware DOM injection with service workers and Chrome storage APIs for persistent performance
- * Deployed the extension, helping 100+ students determine class difficulty

Pill-E – BioEngineering Makeathon | Embedded Systems, IoT | GitHub

Apr. 2026

- * Built the full embedded system for a smart pill dispenser as the sole embedded engineer in a 3-person team
- * Developed control logic for precise pill dispensing and real-time device state management
- * Implemented device-to-mobile communication layer to interface with a companion app built by teammates
- * Debugged and validated hardware performance through iterative testing and calibration of the dispensing system

Smart Surveillance System – ACM Forge | Full-Stack, Computer Vision, IoT | GitHub

Apr. 2026 – Jun. 2026

- * Collaborated in a multidisciplinary team to develop a smart surveillance system with real-time video detection
- * Built a live monitoring dashboard using Next.js, React, and TypeScript for real-time system visualization
- * Integrated computer vision streaming pipelines through Flask video feeds and MQTT messaging

DevTunes – Google DeepMind Hackathon | AI, Python, Software Design | GitHub

Mar. 2026

- * Worked in a 3-person team to build an AI system that analyzes source code to generate context-aware soundtracks
- * Led project documentation, including system design write-ups, workflow diagrams, and technical specifications
- * Contributed to designing modular workflows for code parsing, scene detection, and automated prompt generation